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UniSec Medium Voltage Switchgear PT MULTIMAS NABATI ASAHAN MVD-10&MVD-11 20kV Centralized MVD Room SCHEMATIC DIAGRAM WBS BUS TIE H09															
				Date Prep. 15.11.2018		PROJECT TITLE		ABB EPMV		COVER SHEET		Scale		= H09	
				Prepared YR		MV Switchgear for MNA Serang Plant				MVD-10&MVD-11 20kV Centralized MVD Room		Customer PT Multimas Nabati Asahan		& 303	
00		Issued for Approval		02.01.19		YR		Checked AW		SCHEMATIC DIAGRAM		Drawing Nbr. IDUNISEC-BE318042-301		Sh. 1	
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UniSec

Medium Voltage Switchgear

PT MULTIMAS NABATI ASAHAN

MVD-10&MVD-11 20kV Centralized MVD Room

SCHEMATIC DIAGRAM

WBS BUS TIE

H09

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			Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	Index of Sheet MVD-10&MVD-11 20kV Centralized MVD Room SCHEMATIC DIAGRAM WBS BUS TIE	Scale		= H09			
			Prepared	YR					Customer	PT Multimas Nabati Asahan	& 303			
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A	REFERENCE DESIGNATION OF OBJECTS IN ELECTRICAL DOCUMENTS				-BGB5		POSITION SWITCH OF CIRCUIT-BREAKER SIGNALLING UNDERVOLTAGE RELEASE ENERGIZED		A						
	(IN COMPLIANCE WITH STANDARD IEC 81346-2 AND ABB TECHNICAL STANDARD 2NBA000001)				-BGB6		POSITION SWITCH OF CIRCUIT-BREAKER SIGNALLING UNDERVOLTAGE RELEASE EXCLUDED MECHANICALLY								
B	DESIGNATION		DESCRIPTION		-BGB7		POSITION SWITCHES OF CIRCUIT-BREAKER ON SWITCHGEAR		B						
	-AA		MULTIFUNCTION UNIT (CENTRAL UNIT)		-BGB8...-BGB10		POSITION SWITCHES OF CIRCUIT-BREAKER								
	-BAD1,-BAD4		CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L1		-BGB11		POSITION SWITCH OF CIRCUIT-BREAKER OPERATING FOR MANUAL OPENING								
	-BAD2,-BAD5		CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L2		-BGB12		PROXIMITY SWITCH SIGNALLING CIRCUIT-BREAKER IN OPEN POSITION								
	-BAD3,-BAD6		CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L3		-BGB13		PROXIMITY SWITCH SIGNALLING CIRCUIT-BREAKER IN CLOSED POSITION								
	-BAR		VOLTAGE PROTECTION RELAY		-BGD		POSITION SWITCH OF L.V. INSTRUMENT COMPARTMENT DOOR.								
	-BAS1		VOLTAGE SENSOR LOCATED ON PHASE L1		-BGD1		POSITION SWITCH OF CIRCUIT-BREAKER COMPARTMENT DOOR								
	-BAS2		VOLTAGE SENSOR LOCATED ON PHASE L2		-BGD2		POSITION SWITCH OF CABLE COMPARTMENT DOOR								
	-BAS3		VOLTAGE SENSOR LOCATED ON PHASE L3		-BGE1		POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE IN OPEN POSITION								
	-BAT1		VOLTAGE TRANSFORMER LOCATED ON PHASE L1		-BGE2		POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE IN CLOSED POSITION								
C	-BAT2		VOLTAGE TRANSFORMER LOCATED ON PHASE L2		-BGE3		POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)		C						
	-BAT3		VOLTAGE TRANSFORMER LOCATED ON PHASE L3		-BGE4		POSITION SWITCHES OF OPERATION OF EARTHING SWITCH -QCE ACTUATED BY ELECTRIC MOTOR								
	-BAT4		VOLTAGE TRANSFORMER IN BUSBAR COMPARTMENT LOCATED ON PHASE L1		-BGE5		POSITION SWITCHES OF OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2 ACTUATED BY ELECTRIC MOTOR								
	-BAT5		VOLTAGE TRANSFORMER IN BUSBAR COMPARTMENT LOCATED ON PHASE L2		-BGE7		POSITION SWITCH SIGNALLING EARTHING SWITCH NOT IN MANUAL OPERATION								
	-BAT6		VOLTAGE TRANSFORMER IN BUSBAR COMPARTMENT LOCATED ON PHASE L3		-BGE8		POSITION SWITCHES OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH- DISCONNECTOR -QBS								
	-BCD		DIFFERENTIAL PROTECTIVE RELAY		-BGE11		POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE1 OR -QCE2 IN OPEN POSITION								
	-BCF		FEEDER PROTECTION RELAY		-BGE12		POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE1 OR -QCE2 IN CLOSED POSITION								
	-BCG		GENERATOR PROTECTION RELAY		-BGE13		POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE1 OR -QCE2 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)								
	-BCM		MOTOR PROTECTION RELAY		-BGE14, -BGE15		POSITION SWITCHES FOR OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS, ACTUATED BY ELECTRIC MOTOR								
	-BCN		NEUTRAL (RESIDUAL) CURRENT TRANSFORMER		-BGF		POSITION SWITCHES OF MEDIUM VOLTAGE FUSES								
D	-BCP		TRANSFORMER PROTECTION RELAY		-BGI1		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD IN OPEN POSITION		D						
	-BCR		CURRENT PROTECTION RELAY		-BGI2		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD IN CLOSED POSITION								
	-BCS1		CURRENT SENSOR LOCATED ON PHASE L1		-BGI3		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)								
	-BCS2		CURRENT SENSOR LOCATED ON PHASE L2		-BGI4		POSITION SWITCHES OF OPERATION OF DISCONNECTOR -QBD1 ACTUATED BY ELECTRIC MOTOR								
	-BCS3		CURRENT SENSOR LOCATED ON PHASE L3		-BGI5		POSITION SWITCHES OF OPERATION OF DISCONNECTOR -QBD2 ACTUATED BY ELECTRIC MOTOR								
	-BCT1		CURRENT TRANSFORMER LOCATED ON PHASE L1		-BGI6		POSITION SWITCH SIGNALLING DISCONNECTOR (OR SWITCH-DISCONNECTOR) -QBD NOT IN EMERGENCY MANUAL OPERATION								
	-BCT2		CURRENT TRANSFORMER LOCATED ON PHASE L2		-BGI7		POSITION SWITCH SIGNALLING DICONNECTOR (OR SWITCH-DISCONNECTOR) -QBD NOT IN MANUAL OPERATION								
	-BCT3		CURRENT TRANSFORMER LOCATED ON PHASE L3		-BGI8		POSITION SWITCHES OF SWITCH-DISCONNECTOR -QBS								
	-BCT4,-BCT7		CURRENT TRANSFORMER LOCATED ON PHASE L1		-BGI9		POSITION SWITCHES OF OPERATION OF SWITCH-DISCONNECTOR -QBS ACTUATED BY ELECTRIC MOTOR								
	-BCT5,-BCT8		CURRENT TRANSFORMER LOCATED ON PHASE L2		-BGI11		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD1 IN OPEN POSITION								
E	-BCT6,-BCT9		CURRENT TRANSFORMER LOCATED ON PHASE L3		-BGI12		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD1 IN CLOSED POSITION		E						
	-BCZ		DISTANCE PROTECTION RELAY		-BGI13		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD1 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)								
	-BEF		FREQUENCY PROTECTION RELAY		-BGI14, -BGI15		POSITION SWITCHES OF OPERATION OF DISCONNECTOR -QBD1 ACTUATED BY ELECTRIC MOTOR								
	-BER		SUPERVISION RELAYS		-BGI21		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD2 IN OPEN POSITION								
	-BES		SYNCHRONIZING RELAY		-BGI22		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD2 IN CLOSED POSITION								
	-BET		THERMAL PROTECTION RELAY		-BGI23		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD2 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)								
	-BGA1		POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN CIRCUIT-BREAKER COMPARTMENT		-BGI24, -BGI25		POSITION SWITCHES OF OPERATION OF DISCONNECTOR -QBD2 ACTUATED BY ELECTRIC MOTOR								
	-BGA2		POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN BUSBAR COMPARTMENT (SINGLE BUSBAR SYSTEM) OR IN FRONT BUSBAR COMPARTMENT (DOUBLE BUSBAR SYSTEM)												
	-BGA3		POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN CABLE COMPARTMENT												
	-BGA4		POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN REAR BUSBAR COMPARTMENT (DOUBLE BUSBAR SYSTEM)												
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					Date Prep.	15.11.2018	PROJECT TITLE		ABB	EPMV	REFERENCE DESIGNATIONS		Scale		= H09
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A	-BGI31	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION						-CA	CAPACITORS						A		
	-BGI32	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN CLOSED POSITION						-CA1	CAPACITOR FOR FRONT VENTILATOR							B	
	-BGI33	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)						-CA2	CAPACITOR FOR REAR VENTILATOR								C
	-BGI41...	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN FEEDER POSITION						-EA1	LIGHTING LAMP LOCATED IN L.V. INSTRUMENT COMPARTMENT.								
-BGI42	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS IN OPEN POSITION						-EA2	LIGHTING LAMP LOCATED IN CABLE COMPARTMENT						E			
-BGI43...	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN EARTH POSITION						-EA3	LIGHTING LAMP LOCATED IN OPERATING MECHANISM ENCLOSURE							F		
-BGI51	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN FEEDER POSITION						-EA4	LIGHTING LAMP LOCATED IN CIRCUIT BREAKER COMPARTIMENT								A	
-BGI52	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 IN OPEN POSITION						-EB1	HEATER LOCATED IN CABLE COMPARTMENT									B
-BGI53	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN EARTH POSITION						-EB2	HEATER LOCATED IN CIRCUIT-BREAKER COMPARTMENT						C			
-BGK	POSITION SWITCH OF KEY LOCK						-EB3	HEATER LOCATED IN MOTOR CABINET							D		
-BGL1	POSITION SWITCH OF ELECTROMECHANICAL LOCK -RLE1						-EB4	HEATER FOR VENTILATION TEST								E	
-BGS1...-BGS4	POSITION SWITCHES OF CIRCUIT-BREAKER SPRINGS						-EB5	HEATER LOCATED IN L.V. INSTRUMET COMPARTMENT.									F
-BGS5	PROXIMITY SWITCH SIGNALLING SPRINGS IN CHARGED POSITION						-EB7	HEATERS LOCATED IN LEFT SIDE OF THE OPERATING MECHANISM ENCLOSURE						A			
-BGT1	POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN SERVICE POSITION						-EB8	HEATERS LOCATED IN RIGHT SIDE OF THE OPERATING MECHANISM ENCLOSURE							B		
-BGT2	POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN TEST POSITION						-FA1	SURGE ARRESTER LOCATED ON PHASE L1								C	
-BGT3	POSITION SWITCH ON TRUCK SIGNALLING TRUCK NOT IN ISOLATING TRAVEL POSITION						-FA2	SURGE ARRESTER LOCATED ON PHASE L2									D
-BGT4	POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN SERVICE POSITION						-FA3	SURGE ARRESTER LOCATED ON PHASE L3						E			
-BGT5	POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN TEST POSITION						-FCD	FUSE-DISCONNECTORS FOR PROTECTION OF AUXILIARY CIRCUITS							F		
-BGT6	POSITION SWITCHES SIGNALLING V.T. TRUCK IN SERVICE POSITION						-FCF1	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L1								A	
-BGT7	POSITION SWITCHES SIGNALLING V.T. TRUCK IN TEST POSITION						-FCF2	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L2									B
-BGT8	POSITION SWITCHES SIGNALLING EARTHING TRUCK IN SERVICE POSITION						-FCF3	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L3						C			
-BGT11	PROXIMITY SWITCH SIGNALLING TRUCK IN SERVICE POSITION						-FCM1	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF SPRINGS CHARGING MOTOR ON MAIN CIRCUIT-BREAKER							D		
-BGT12	PROXIMITY SWITCH SIGNALLING TRUCK IN TEST POSITION						-FCM2...-FCM10	MINIATURE CIRCUIT-BREAKERS FOR PROTECTION OF AUXILIARY CIRCUITS								E	
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-BPA6	PRESSURE SWITCH LOCATED IN BUSBAR OR IN BUSBAR AND CIRCUIT-BREAKER COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GQ1	FRONT VENTILATOR						E			
-BPA7	PRESSURE SWITCH LOCATED IN VOLTAGE TRANSFORMER ISOLATING COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GQ2	REAR VENTILATOR							F		
-BPS1	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER OR ON POLE OF PHASE L1 OF CIRCUIT-BREAKER						-GQ3	VENTILATOR LOCATED ON CIRCUIT-BREAKER TRUCK								A	
-BPS2	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L2						-KFA1	AUXILIARY RELAY SIGNALLING LOW GAS PRESSURE									B
-BPS3	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L3						-KFA2	AUXILIARY RELAY SIGNALLING INSUFFICIENT GAS PRESSURE						C			
-BR	FLAME DETECTORS, SMOKE DETECTORS						-KFA3...-KFA9	AUXILIARY RELAYS OR CONTACTORS							D		
-BT	THERMOSTAT						-KFC	CLOSING RELAYS OR CONTACTORS								E	
-BT1	THERMOSTAT FOR DE-ENERGIZATION OF VENTILATORS						-KFI	INTEGRATED CIRCUITS									F
-BT2	THERMOSTAT FOR ENERGIZATION OF VENTILATORS						-KFL	LOCKOUT RELAY						A			
-BT3	THERMOSTAT SIGNALLING HIGH TEMPERATURE ALARM						-KFM	MICROPROCESSORS							B		
-BU	BUCHHOLZ RELAY						-KFN	ANTIPUMPING RELAYS								C	
-BX1	UNIT WITH SENSORS FOR DETECTION OF INTERNAL ARCING						-KFO	OPENING RELAYS OR CONTACTORS									D
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							-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER							D		
							-MAD	MOTOR FOR ELECTRICAL OPERATION OF DISCONNECTOR -QBD								E	
							-MAD1	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD1									F
							-MAD2	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD2						A			
							-MAD9	MOTOR FOR ELECTRICAL OPERATION OF SWITCH-DISCONNECTOR -QBS							B		
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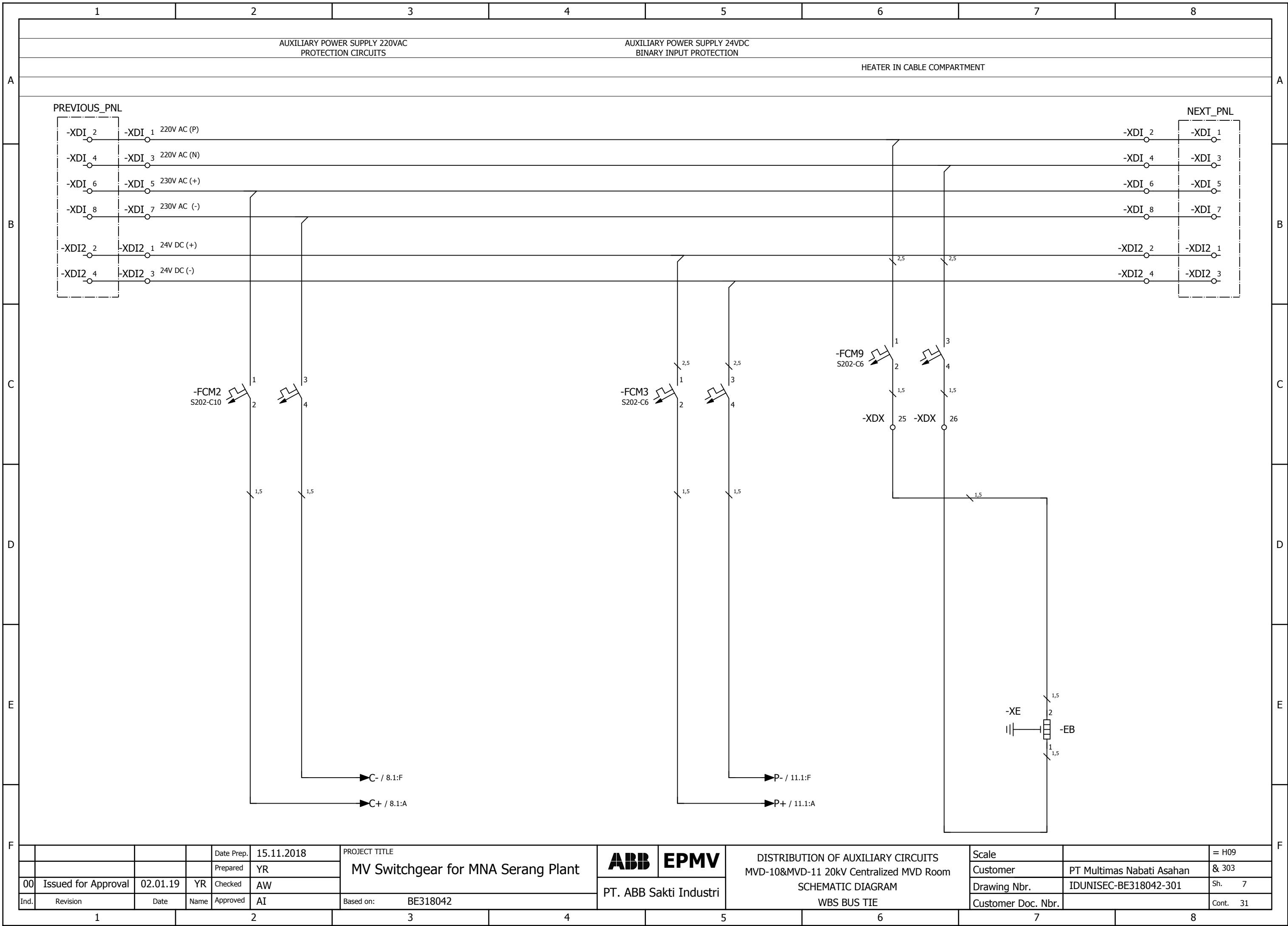
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A	-MAE		MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE		-QBH		MANUAL CIRCUIT-BREAKERS								
	-MAE1		MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2		-QBL		LINKS								
	-MAE8		MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS		-QBM		MINIATURE SWITCH-DISCONNECTORS								
					-QBS		SWITCH-DISCONNECTORS								
B	-MAS		MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING		-QBT		TRUCK								
	-MAT		MOTOR FOR ELECTRICAL OPERATION OF TRUCK RACKING-IN/OUT		-QCA		EARTHING SWITCH FOR ELIMINATION OF INTERNAL ARCING								
	-MBC		CLOSING RELEASE OF CIRCUIT-BREAKER		-QCE		EARTHING SWITCH								
	-MBC4		CLOSING RELEASE OF SWITCH-DISCONNECTOR -QBS		-QCE1		EARTHING SWITCH OF BUSBARS (SINGLE BUSBAR SYSTEM) OR OF FRONT BUSBARS (DOUBLE BUSBAR SYSTEM)								
	-MBO1		FIRST OPENING RELEASE OF CIRCUIT-BREAKER												
	-MBO2		SECOND OPENING RELEASE OF CIRCUIT-BREAKER		-QCE2		EARTHING SWITCH OF REAR BUSBARS (DOUBLE BUSBAR SYSTEM)								
C	-MBO3		OPENING SOLENOID FOR OVERCURRENT RELEASE OF CIRCUIT-BREAKER OR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER		-QQ		CLUTCHES								
					-RAA		ANTIFERRORESONANCE RESISTOR								
	-MBO4		OPENING RELEASE OF SWITCH-DISCONNECTOR -QBS		-RAD		DIODES								
	-MBU		UNDERVOLTAGE RELEASE OF CIRCUIT-BREAKER		-RAI		INDUCTORS								
	-PFB		BLUE SIGNAL LAMPS		-RAR		RESISTORS								
	-PFF		FLAG RELAYS		-RF		FILTERS								
D	-PFG		GREEN SIGNAL LAMPS		-RLE1		ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING								
	-PFR		RED SIGNAL LAMPS		-RLE2		ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT								
	-PFS		SHORT CIRCUIT INDICATORS		-RLE3		ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE								
	-PFV		VOLTAGE INDICATORS												
	-PFV1		VOLTAGE INDICATORS ON FEEDER SIDE		-RLE4		ELECTROMECHANICAL LOCK PREVENTING THE DOOR OPENING OPERATION								
	-PFV2		VOLTAGE INDICATORS ON BUSBAR SIDE		-RLE5		ELECTROMECHANICAL LOCK PREVENTING OPERATION OF DISCONNECTOR -QBD								
E	-PFW		WHITE SIGNAL LAMPS		-RLE6		ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD1								
	-PFX		CROSS INDICATORS, ELECTROMECHANICAL INDICATORS												
	-PFY		YELLOW SIGNAL LAMPS		-RLE7		ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD2								
	-PGA		AMMETERS		-RLE8		ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2								
	-PGC		COUNTERS												
	-PGF		FREQUENCYMETERS		-RLE9		ELECTROMECHANICAL LOCK PREVENTING OPERATION OF SWITCH-DISCONNECTOR -QBS								
F	-PGH		HOURMETERS		-RLE11		ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD4								
	-PGI		PROTECTION AND CONTROL UNIT: HUMAN MACHINE INTERFACE		-RLE12		ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD5								
	-PGI1		HUMAN MACHINE INTERFACE OF ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON CIRCUIT-BREAKER)												
					-SFA		AMMETRIC SWITCHES								
	-PGI2		HUMAN MACHINE INTERFACE FOR ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON SWITCHGEAR)		-SFC		CONTROL SWITCHES, CLOSING PUSH-BUTTONS								
	-PGJ		ACTIVE ENERGY METERS		-SFL		LOCKING CONTACTS								
	-PGK		REACTIVE ENERGY METERS		-SFO		OPENING PUSH-BUTTONS								
	-PGM		MULTIFUNCTION INDICATORS		-SFR		RESET PUSH-BUTTONS								
	-PGP		POWER-FACTOR METERS		-SFS		SELECTOR SWITCHES								
	-PGQ		VARMETERS		-SFT		TEST PUSH-BUTTONS								
E	-PGS		SYNCHRONOSCOPES		-SFU		UNLOCKING PUSH-BUTTONS								
	-PGV		VOLTMETERS		-SFV		VOLTMETRIC SWITCHES								
	-PGW		WATTMETERS		-TA		POWER TRANSFORMERS								
	-PJ		ACOUSTICAL SIGNAL DEVICES (BELLS, SIRENS)		-TB		CONVERTER								
	-QAB		CIRCUIT-BREAKERS		-TB1		RECTIFIER FOR FIRST OPENING RELEASE								
	-QAC		CONTACTORS (FOR POWER)		-TB2		RECTIFIER FOR SECOND OPENING RELEASE								
	-QBD		DISCONNECTORS		-TB3		RECTIFIER FOR CLOSING RELEASE								
	-QBD1		FRONT BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)		-TB4		RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING								
	-QBD2		REAR BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)		-TB5		RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT								
	-QBD3		FEEDER DISCONNECTORS (DOUBLE BUSBAR SYSTEM)		-TB6		RECTIFIER FOR UNDERVOLTAGE RELEASE								
	-QBD4		TWO-WAY DISCONNECTOR		-TB7		ERECTIFIER FOR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER								
	-QBD5		TWO-WAY DISCONNECTOR ON BUSBAR RISER												
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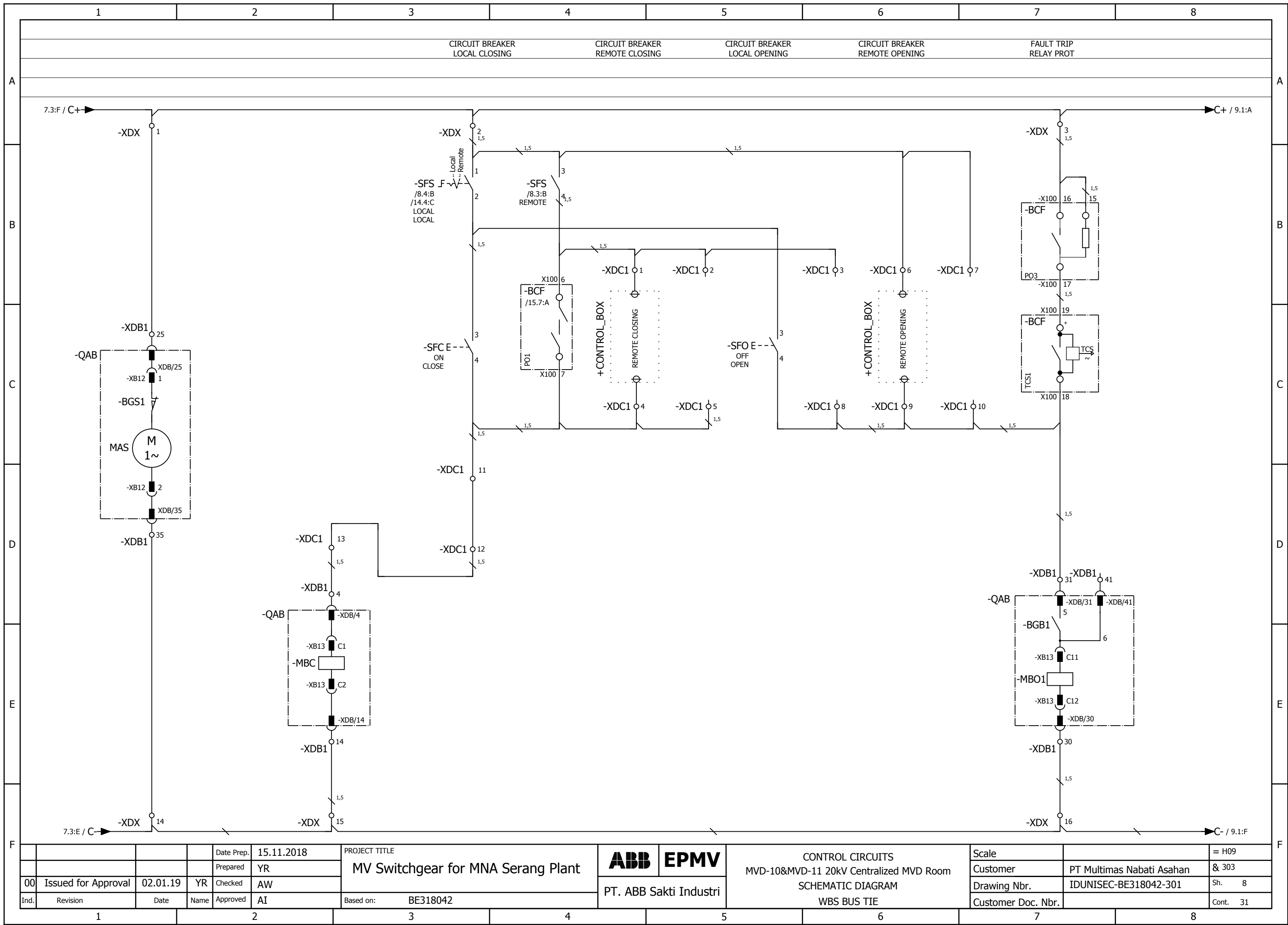
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A	-TFA		ACTIVE POWER TRANSDUCERS		-XD11	TERMINAL BLOCK FOR INTERCONNECTIONS OF CONTROL CIRCUITS		A							
	-TFC		CURRENT TRANSDUCERS			-XD12			TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING						
	-TFF		FREQUENCY TRANSDUCERS		-XD13	TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR SWITCH-DISCONNECTOR OPERATION									
	-TFJ		ACTIVE ENERGY TRANSDUCERS			-XD14			TERMINAL BLOCK FOR INTERCONNECTIONS OF A.C. AUXILIARY CIRCUITS						
	-TFK		REACTIVE ENERGY TRANSDUCERS		-XD16	TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS) OF VOLTAGE CIRCUITS									
	-TFM		MULTIFUNCTION TRANSDUCERS			-XD17			TERMINAL BLOCK FOR INTERCONNECTION OF MOD-BUS						
	-TFP		POWER-FACTOR TRANSDUCERS		-XDJ	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF DISCONNECTORS -QBD									
	-TFQ		REACTIVE POWER TRANSDUCERS			-XDJ1			TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR -QBS						
B	-TFS		SIGNAL CONVERTERS		-XDM	SEALABLE TERMINAL BLOCK FOR MEASUREMENT		B							
	-TFV		VOLTAGE TRANSDUCERS			-XDS			CURRENT SOCKETS						
	-WA		BUSBARS		-XDT	TERMINAL BLOCK FOR POSITION CONTACTS OF TRUCK AND OF CONNECTOR -XDB									
	-WBC		POWER CABLES			-XDU			CONNECTOR FOR ISOLATION OF CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK						
	-WE		EARTHING CABLES FOR L.V. INSTRUMENT COMPARTMENT.		-XDU1	TERMINAL BLOCK FOR CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK									
	-WGC		CONTROL CABLES			-XDV			TERMINAL BLOCK FOR CIRCUITS OF VOLTAGE TRANSFORMERS						
	-WGS		COMMUNICATION CABLES (SHIELDED TWISTED PAIRS)		-XDV1	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L1									
	-WH		OPTICAL FIBRES			-XDV2			CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L2						
	-XDA		TERMINAL BLOCK FOR CIRCUITS OF CURRENT TRANSFORMERS		-XDV3	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L3									
	-XDA1		CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L1			-XDV4			CONNECTOR FOR CIRCUITS OF ANTIFERRORESONANCE RESISTOR						
	-XDA2		CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L2		-XDX	SUPPORT TERMINAL BLOCK									
	-XDA3		CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L3			-XE1			EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, LEFT SIDE						
C	-XDA5		CONNECTOR FOR CIRCUITS OF NEUTRAL (RESIDUAL) CURRENT TRANSFORMER		-XE2	EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, RIGHT SIDE									
	-XDB		CONNECTOR FOR ISOLATION OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK) AUXILIARY CIRCUITS			-XE5			EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, VOLTAGE INDICATOR SIDE						
	-XDB1		CONNECTOR FOR AUXILIARY CIRCUITS OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK)		-XE6	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, REAR BOTTOM SIDE									
	-XDB2		CONNECTOR FOR ADDITIONAL AUXILIARY CIRCUITS OF CIRCUIT-BREAKER			-XE7			EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, LEFT SIDE						
	-XDB10...-XDB89		CONNECTOR FOR CIRCUIT-BREAKER INTERNAL CIRCUITS		-XE8	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, FRONT BOTTOM SIDE									
	-XDB7		CONNECTOR FOR TRUCK POSITION CONTACTS			-XE9			EARTHING TERMINAL BLOCK FOR TRANSFORMERS IN BUSBAR COMPARTMENT						
	-XDB91		CONNECTOR FOR OPENING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR		-XDC	CUSTOMER TERMINAL BLOCK									
	-XDB92		CONNECTOR FOR CLOSING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR			-XDC1			CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF CIRCUIT BREAKER						
D	-XDB93		CONNECTOR FOR POSITION SWITCHES OF SWITCH-DISCONNECTOR		-XDC2	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR									
	-XDB95		CONNECTOR FOR CIRCUITS OF EARTHING SWITCH LOCKING MAGNET			-XDC3			CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF PROTECTION RELAY AND MULTIFUNCTION UNIT						
	-XDC		CUSTOMER TERMINAL BLOCK		-XDC4	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF VOLTAGE TRANSFORMERS									
	-XDC1		CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF CIRCUIT BREAKER			-XDC5			CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF MINIATURE CIRCUIT BREAKER						
	-XDC2		CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR		-XDE	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE									
	-XDC3		CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF PROTECTION RELAY AND MULTIFUNCTION UNIT			-XDE1			TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE						
	-XDC4		CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF VOLTAGE TRANSFORMERS		-XDE2	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2									
	-XDC5		CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF MINIATURE CIRCUIT BREAKER			-XDE3			TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2						
E	-XDE		TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE		-XDF	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF FORCED COOLING VENTILATORS									
	-XDE1		TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE			-XDF1			CONNECTOR FOR ISOLATION OF CIRCUITS OF FRONT VENTILATOR						
	-XDE2		TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2		-XDF2	CONNECTOR FOR REAR VENTILATOR CIRCUITS									
	-XDE3		TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2			-XDF3			CONNECTOR FOR FRONT VENTILATOR CIRCUITS						
	-XDF		TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF FORCED COOLING VENTILATORS		-XDH	TERMINAL BLOCK FOR VARIOUS CIRCUITS (HEATERS, POSITION SWITCHES DETECTING INTERNAL ARCING, ETC.)									
	-XDF1		CONNECTOR FOR ISOLATION OF CIRCUITS OF FRONT VENTILATOR			-XDI			TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS)						
	-XDF2		CONNECTOR FOR REAR VENTILATOR CIRCUITS		-XDI	TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS)									
	-XDF3		CONNECTOR FOR FRONT VENTILATOR CIRCUITS												
F	-XDH		TERMINAL BLOCK FOR VARIOUS CIRCUITS (HEATERS, POSITION SWITCHES DETECTING INTERNAL ARCING, ETC.)						F						
	-XDI		TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS)												
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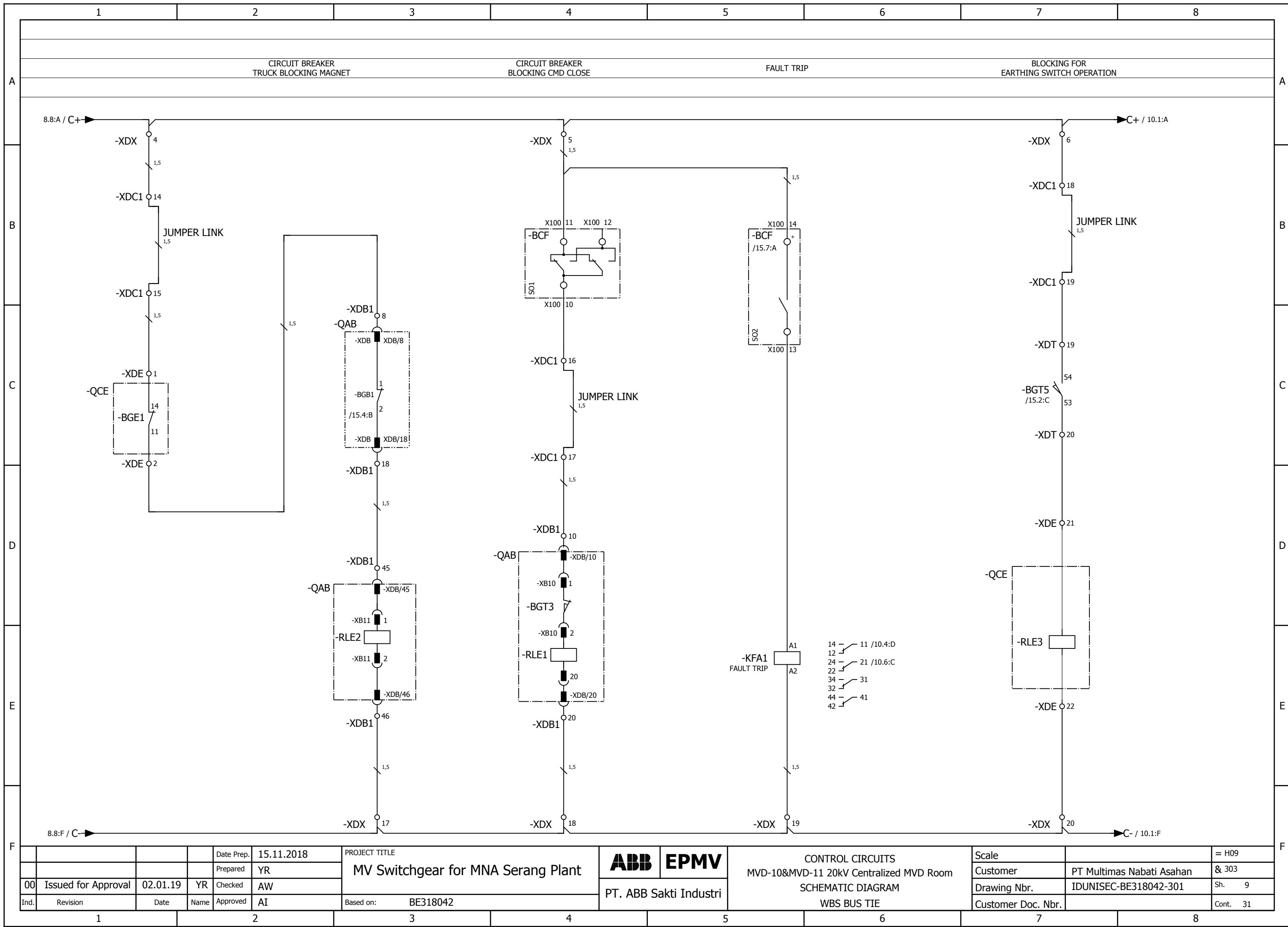
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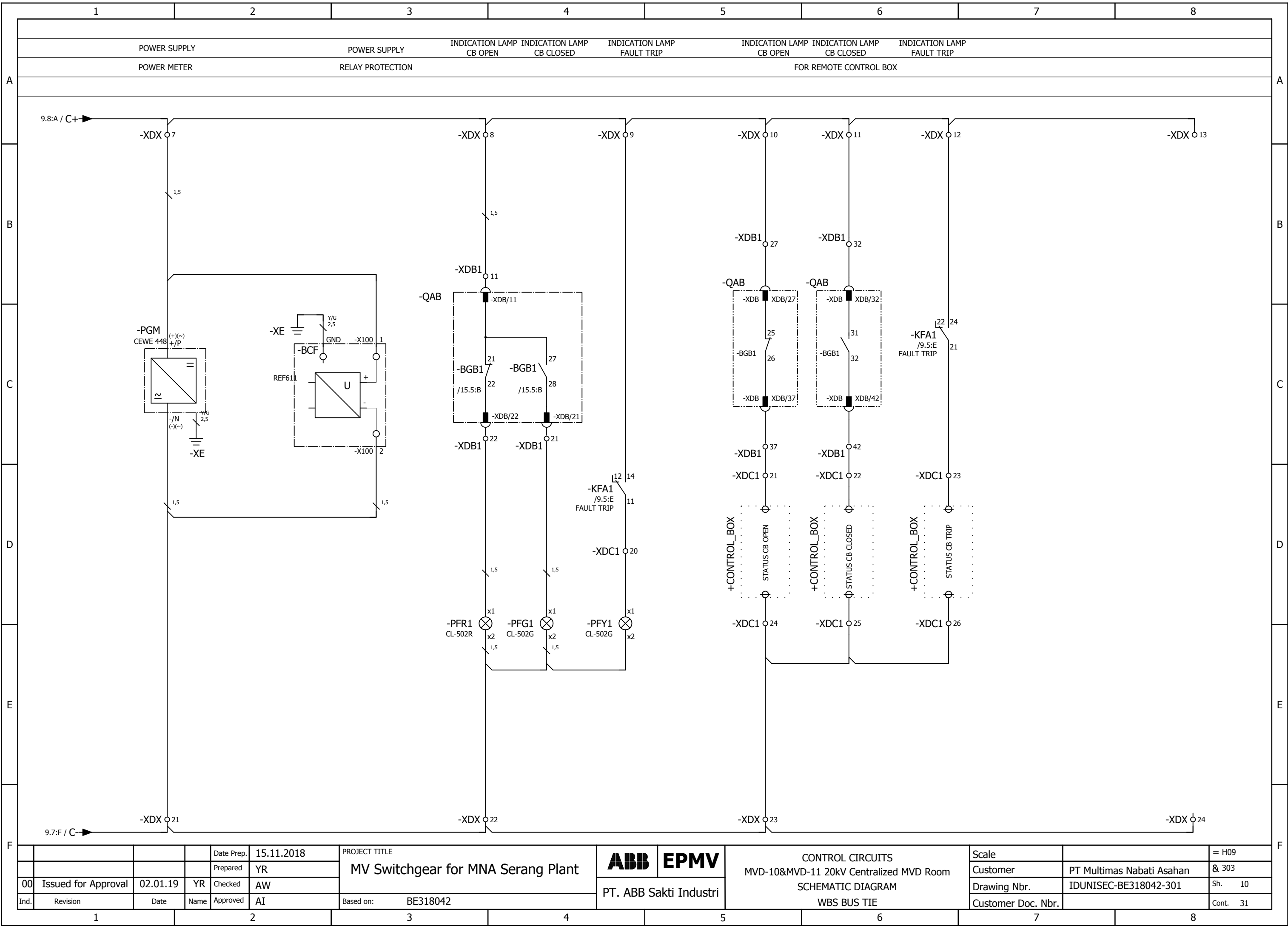
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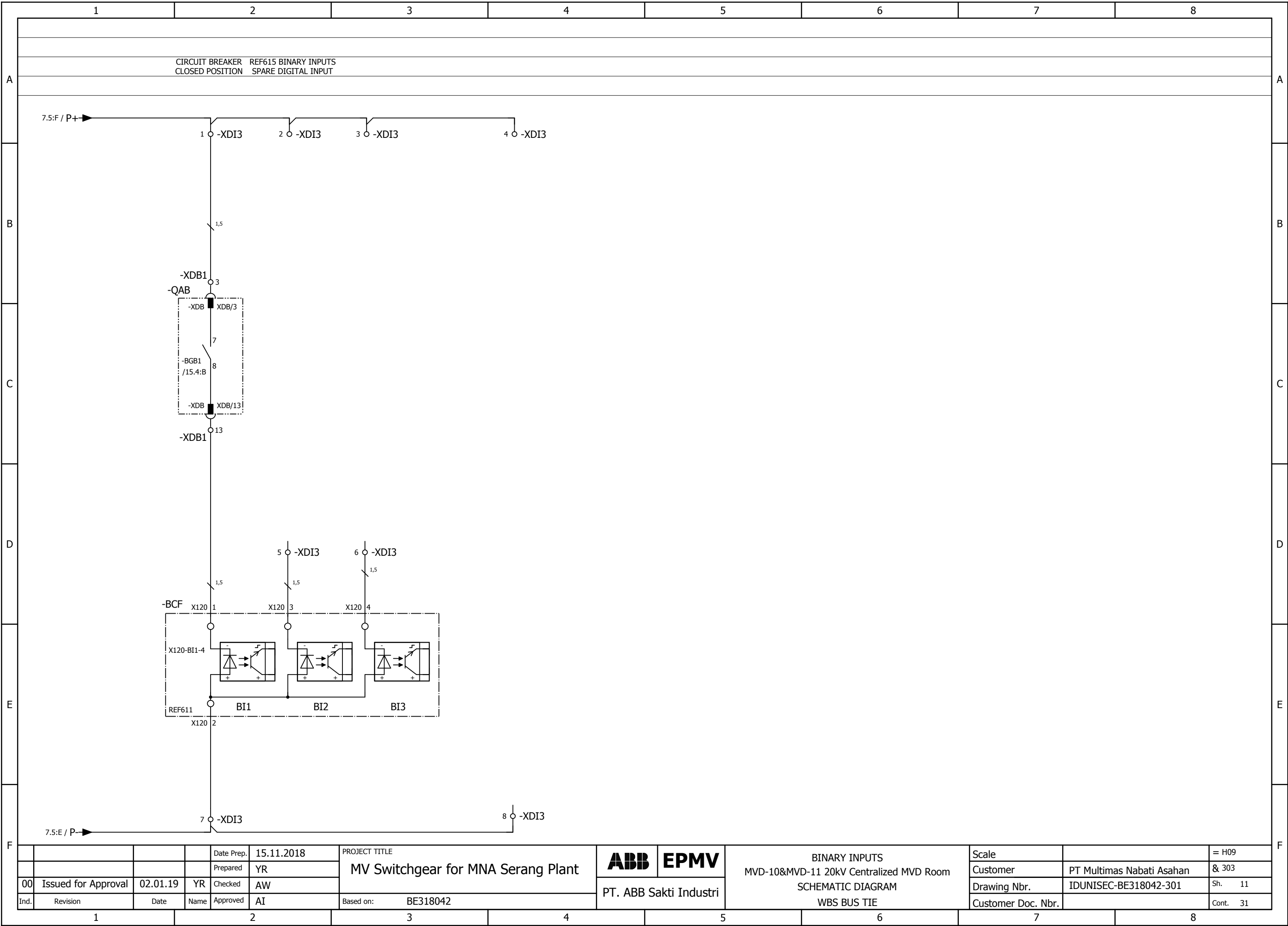
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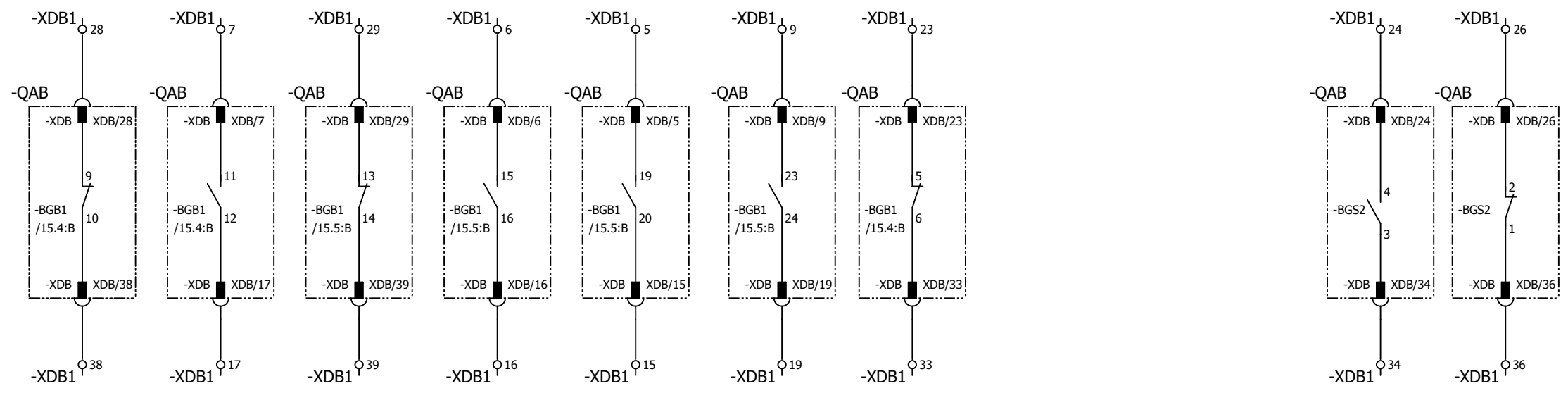
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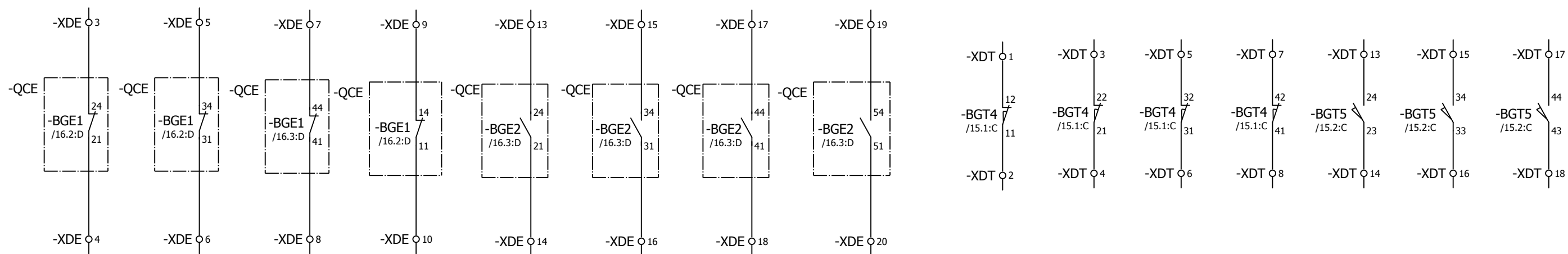
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				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	BINARY INPUTS MVD-10&MVD-11 20kV Centralized MVD Room SCHEMATIC DIAGRAM WBS BUS TIE	Scale		= H09
				Prepared	YR					Customer	PT Multimas Nabati Asahan	& 303
00	Issued for Approval	02.01.19	YR	Checked	AW		Drawing Nbr.	IDUNISEC-BE318042-301		Sh. 11		
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	PT. ABB Sakti Industri			Customer Doc. Nbr.		Cont. 31

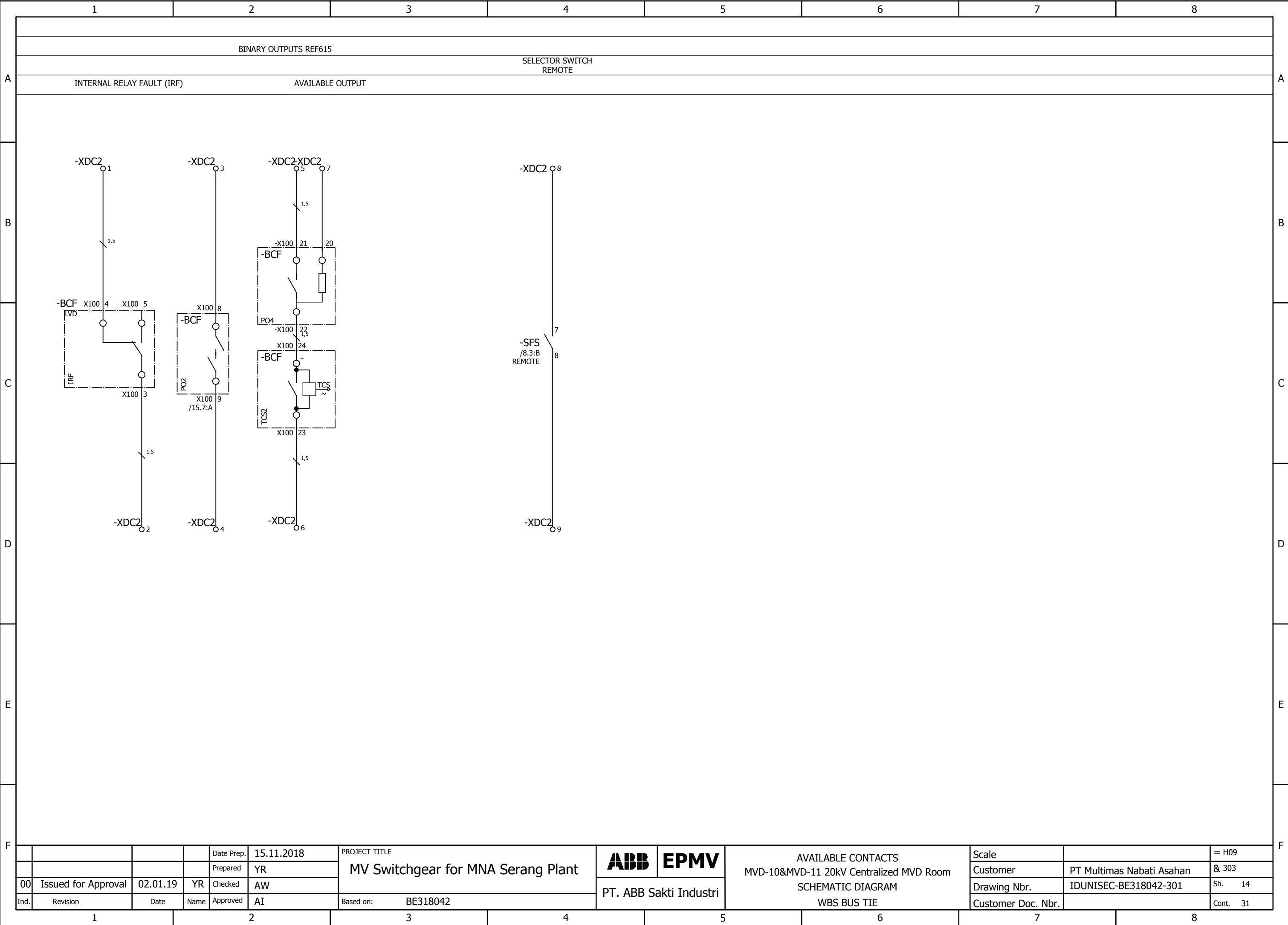


				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant			AVAILABLE CONTACTS MVD-10&MVD-11 20kV Centralized MVD Room SCHEMATIC DIAGRAM WBS BUS TIE	Scale		= H09
				Prepared	YR					Customer	PT Multimas Nabati Asahan	& 303
00	Issued for Approval	02.01.19	YR	Checked	AW		PT. ABB Sakti Industri	Drawing Nbr.		IDUNISEC-BE318042-301	Sh. 12	
Ind.	Revision	Date	Name	Approved	AI			Based on: BE318042		Customer Doc. Nbr.		Cont. 31



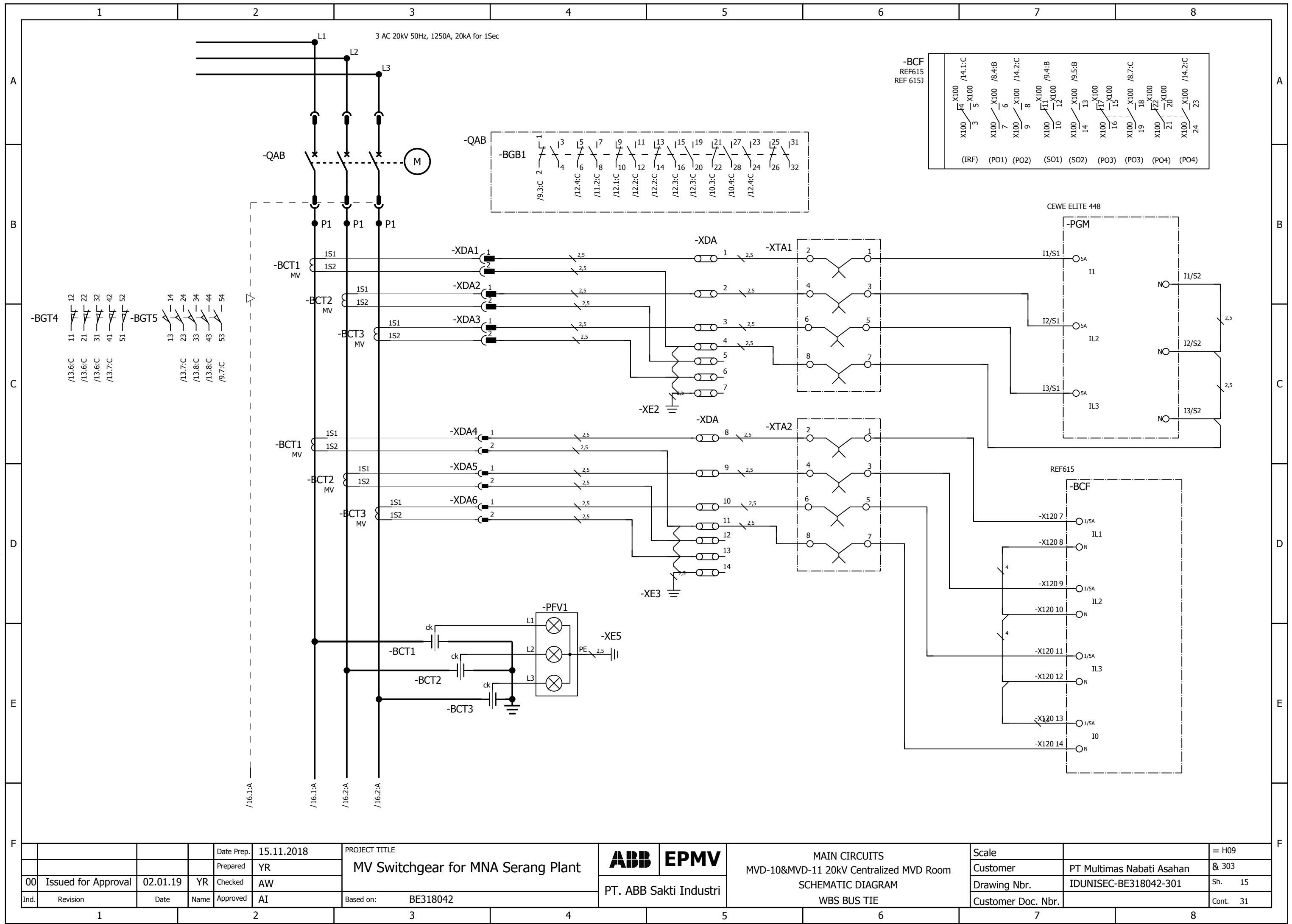
				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	AVAILABLE CONTACTS MVD-10&MVD-11 20kV Centralized MVD Room SCHEMATIC DIAGRAM WBS BUS TIE	Scale		= H09
				Prepared	YR					PT. ABB Sakti Industri	Customer	PT Multimas Nabati Asahan
00	Issued for Approval	02.01.19	YR	Checked	AW		Drawing Nbr.	IDUNISEC-BE318042-301			Sh.	13
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	Customer Doc. Nbr.			Cont.	31	

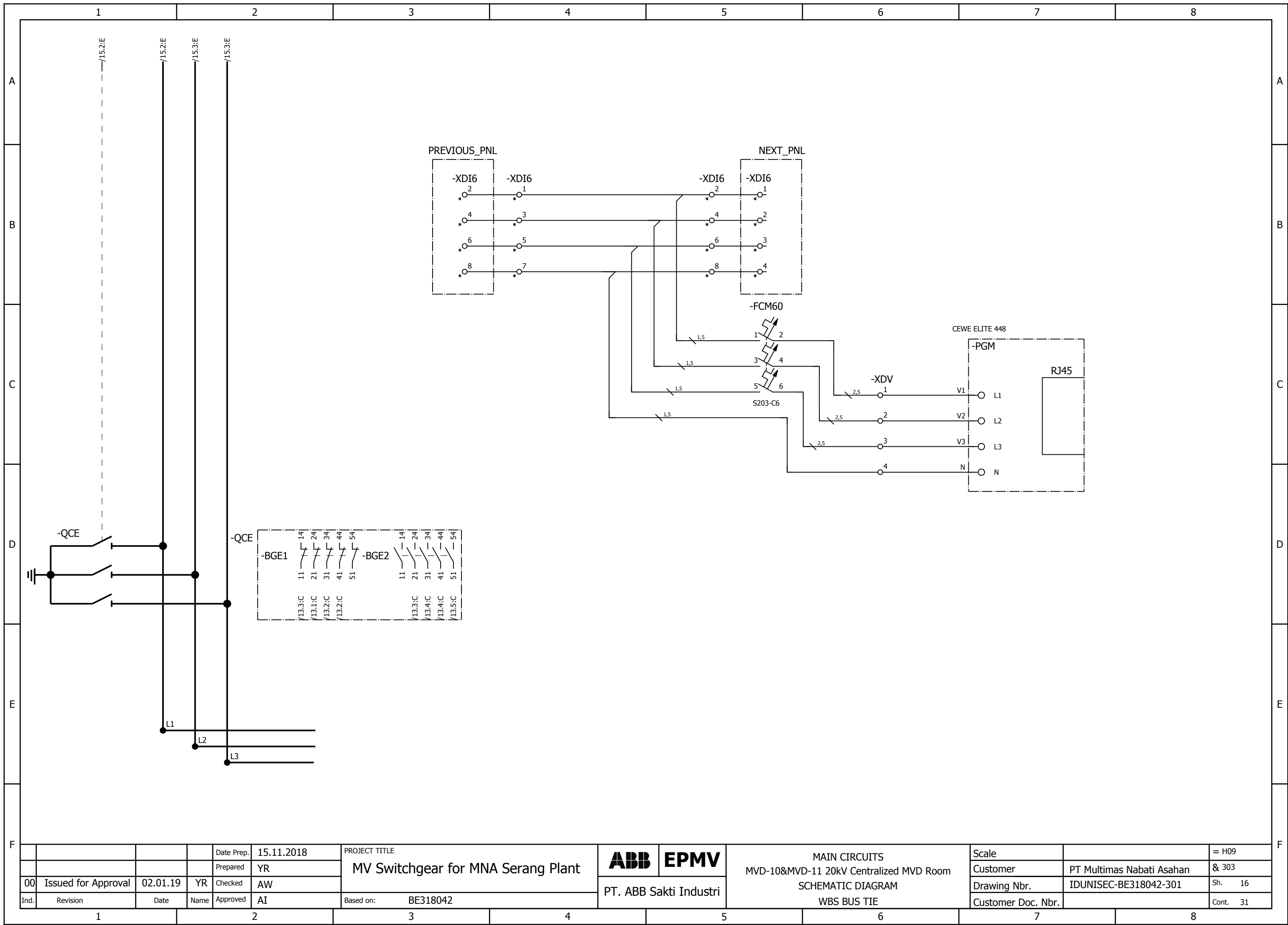
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00	Issued for Approval		02.01.19	YR	Checked	AW	Based on:	BE318042	PT. ABB Sakti Industri	AVAILABLE CONTACTS MVD-10&MVD-11 20kV Centralized MVD Room SCHEMATIC DIAGRAM WBS BUS TIE	Scale		= H09	
	Revision		Date	Name	Approved	AI					Customer	PT Multimas Nabati Asahan		& 303
											Drawing Nbr.	IDUNISEC-BE318042-301		Sh. 14
											Customer Doc. Nbr.			Cont. 31

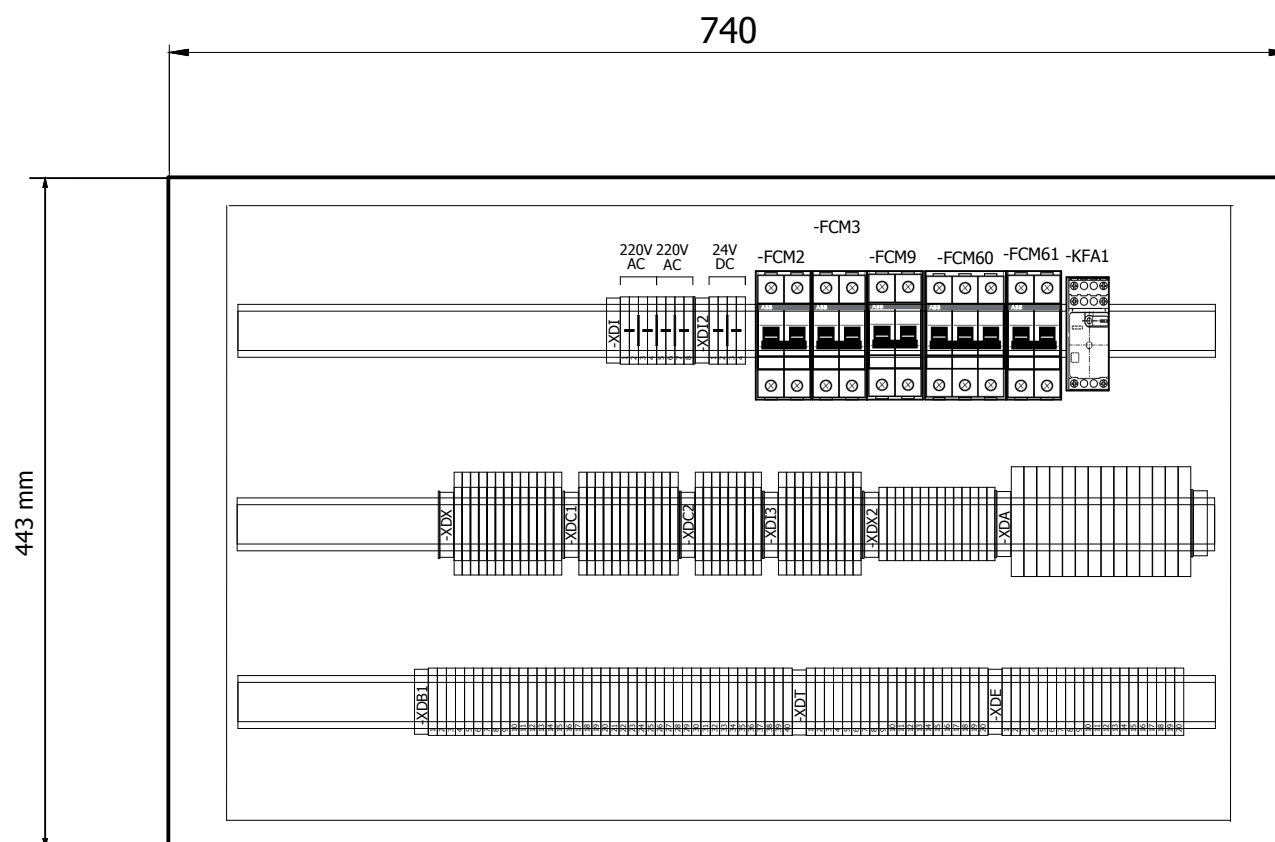
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No.	EQUIPMENT DESCRIPTION	ENGRAVING
1	-SFS (LOCAL-OFF-REMOTE SELECTOR SWITCH, K&N)	-SFS, LOCAL-OFF-REMOTE
2	-SFC (PUSH BUTTON CB CLOSE, GREEN, ABB)	-SFC, CB CLOSE
3	-SFO (PUSH BUTTON CB OPEN, RED, ABB)	-SFO, CB OPEN
4	-PFY1 (LAMP INDICATION FOR CB TRIP, YELLOW, ABB)	-PFY1, CB TRIP
5	-PFR1 (LAMP INDICATION FOR CB CLOSE, RED, ABB)	-PFR1, CB CLOSE
6	-PFG1 (LAMP INDICATION FOR CB OPEN, GREEN, ABB)	-PFG1, CB OPEN
7	-PGJ (DIGITAL MEASUREMENT, CEWE 448)	-PGJ, DIGITAL MEASUREMENT
8	-BCF (FEEDER PROTECTION RELAY, REF615J, ABB)	-BCF, RELAY PROTECTION
9	-XTA1, XTA2 (TEST BLOCK , YONGSUNG)	-XTA, CURRENT TEST BLOCK

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[illegible]

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	LV BOX LAYOUT MVD-10&MVD-11 20kV Centralized MVD Room SCHEMATIC DIAGRAM WBS BUS TIE	Scale		= H09	
				Prepared	YR					PT. ABB Sakti Industri	Customer	PT Multimas Nabati Asahan	& 303
00	Issued for Approval	02.01.19	YR	Checked	AW						Drawing Nbr.	IDUNISEC-BE318042-301	Sh. 18
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	Customer Doc. Nbr.			Cont. 31			

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Summarized parts list															
IDABB_F02_005-IDABB															
Carry over: 0.00															
TAG ID		DESCRIPTION				MANUFACTUR/BRAND		PART NUMBER		QUANTITY		REMARKS			
-BCF		Feeder Management Relay REF611HBAABC1NN11G				ABB		REF611HBAABC1NN11G		1		/10.2:C			
-EB		HEATER 230VAC, 50W				SANYANG		CDXN010033P0009		1		/7.7:E			
-FCM2		MCB S202-C10 MCB. 2P, 10A				ABB		2CDS252001R0104		1		/7.2:C			
-FCM3;-FCM9		MCB S202-C6 MCB. 2P, 6A				ABB		2CDS252001R0064		2		/7.5:C;/7.6:C			
-FCM60		MCB S203-C6 MCB. 3P, 6A				ABB		2CDS253001R0064		1		/16.5:C			
-KFA1		Auxiliary Relay 230AC 4COS				PHOENIX		1YRA3DL266230001		1		/9.5:E			
-PFG1		PILOT LIGHT GREEN #CL-523G#230VAC CL-523G				ABB		1SFA619402R5232		1		/10.4:D			
-PFR1		PILOT LIGHT RED #CL-523R#230VAC CL-523R				ABB		1SFA619402R5231		1		/10.3:D			
-PFY1		PILOT LIGHT YELLOW #CL-523Y#230VAC CL-523Y				ABB		1SFA619402R5233		1		/10.4:D			
-PGM		POWER METER CEWE 448 Class : 0.5s; CT Input : 1A; VT Input : 57.7-240V (L-N); Aux. Voltage: 80-300VAC/DC				CEWE		1YRA4EA0350001		1		/10.1:C			
-SFC		PUSH BUTTON, YW1B-M1E10-G GREEN COLOUR				IDEC		1YRA3DA14700001		1		/8.3:C			
-SFO		PUSH BUTTON, YW1B-M1E10-R RED COLOUR				IDEC		1YRA3DA14700002		1		/8.5:C			
-SFS		LOCAL/REMOTE Selector Switch Type : S0-CA10-A723-600-E; 4NC (for LOCAL Position) & 4NO (for REMOTE Position)				K&N		1YRA3DA17940011		1		/8.3:B			
-XTA1;-XTA2		Current Test Block Rated Current : 10A, 250V; Mounting : Flush Mounting				YONGSUNG		YSCTT-04C		2		/15.5:B;/15.5:C			

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F										E	D		C		B	A												
											CABLE	EQUIPMENT	Wire No	Terminal number	TYPE	WIRE NO.	EQUIPMENT	CABLE	REMARK									
					Date Prep.	15.11.2018				=H09+-XDC1																		
					Prepared	YR																						
00	Issued for Approval			02.01.19	YR	Checked	AW																					
Ind.	Revision			Date	Name	Approved	AI																					

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F		E		D		C		B		A	

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										1	2	3	4	5	6	7	8				
										=H09+-XDV											
										REMARK	CABLE	EQUIPMENT	WIRE NO.	TYPE	Terminal number	Wire No	EQUIPMENT	CABLE	REMARK		
												-PGM:V1		PT 2,5	1			-FCM60:2		2,5 mm²	
												-PGM:V2		PT 2,5	2			-FCM60:4		2,5 mm²	
												-PGM:V3		PT 2,5	3			-FCM60:6		2,5 mm²	
												-PGM:N		PT 2,5	4			-XD16:8		1,5 mm²	

F										E										D										C										B										A									